

Supplementary Material for: Robust Branch-Safe Participation for Sequential Mobile Crowdsensing: Model Uncertainty and Disclosure Floors

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Abstract

This supplement provides technical details supporting the main manuscript: comparator adaptations and validation, finite-model robustness and boundary tests, construction of the data-driven ambiguity set, and reproducibility information. It introduces no additional primary claims and preserves both the exploratory status of the replay analyses and the finite-model scope of the robustness guarantee.

1. Comparator adaptations and validation

For the pointwise-maximal-leakage comparator, let $l_i = \Pr(Y = 1 \mid X = i)$ and $m = \sum_i b_i l_i$, with $0 \leq l_i \leq \alpha \mathbf{1}\{i \in S\}$. The task-channel specialization maximizes report probability subject to the published pointwise leakage constraints $l_i \leq e^\epsilon m$ and $1 - l_i \leq e^\epsilon (1 - m)$ for every state with positive prior mass [1]. Randomized instances are checked against a general linear program. This is a task-channel specialization rather than a reproduction of every component of the original application.

PRIVIC learns a noisy location channel through Blahut–Arimoto and incremental Bayesian updates [2]. Our BA/IBU/GIBU updates are checked against the authors’ public implementation. The task adapter emits a truthful report only when

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the true state is eligible and the sampled location-channel outcome satisfies the task predicate. Because the adapter changes the output semantics, PRIVIC-T does not inherit the original geo-indistinguishability guarantee. Offline fitting is excluded from online gate timing.

The Bayesian-privacy comparator constrains each positive-probability output by $D_{\text{KL}}(b^v||b) \leq \kappa$ [3]. The successful-report divergence is determined by task support, while the silence bound is solved by bisection. It is used as a per-output diagnostic rather than as a cumulative session guarantee.

2. Finite-model robustness and boundary characterization

The hand-grid ambiguity set uses $v \in \{0.05, 0.3, 0.8\}$ and $\alpha \in \{0.3, 0.648, 0.9\}$. Twenty new synthetic seeds, four users per seed, and 48 slots yield 80 trajectories per reported row at a fixed operating point. When the informed attacker’s exact generating model is contained in the set, R-BSP records no local-cap violation in the reported in-set conditions, as expected from the finite-set construction, while retaining about 39.7% of externally completable normal-task opportunities.

Table 1: Finite-model R-BSP stress at $\rho = 0.1$.

Client model	BSP viol.	R-BSP viol.	BSP retention	R-BSP retention
Nominal	0.0%	0.0%	50.9%	39.7%
$\widehat{\alpha} = 0.3$	12.6%	0.0%	54.2%	39.7%
$\widehat{\alpha} = 0.9$	2.6%	0.0%	47.3%	39.7%
$\widehat{v} = 0.05$	2.8%	0.0%	46.9%	39.7%
$\widehat{v} = 0.8$	14.6%	0.0%	55.2%	39.7%

The same ambiguity set is then evaluated against correct attacker models that are not exact members. The outside-set value $\alpha = 0.98$ yields a 6.93% local-cap

violation rate with a cluster-bootstrap 95% interval of 5.70–8.31%. Interior unlisted values yield zero or rare violations in this finite sample. These results support only a finite-model guarantee and are not evidence of robustness to arbitrary distribution shift.

Table 2: Boundary characterization for attacker models not explicitly enumerated in the hand-grid ambiguity set.

Attacker model	BSP viol.	R-BSP viol.	BSP ret.	R-BSP ret.
$\alpha = 0.50$ (interior)	1.15%	0.00%	52.6%	36.3%
$\alpha = 0.98$ (outside)	18.33%	6.93%	50.0%	40.4%
$v = 0.50$ (interior)	0.65%	0.05%	54.4%	36.4%
$v = 0.95$ (outside)	0.65%	0.03%	55.6%	35.4%

3. Data-driven ambiguity construction

The replay ambiguity set is constructed without using the 53 isolated test windows. The 23 development windows define a coarsened reflecting-walk likelihood

$$\Pr(\text{change} \mid i) = vd_i/4,$$

where d_i is the number of in-grid neighbors. Only change versus no-change events enter this likelihood; destination direction and jump distance are not fitted. Two thousand development state-path-group bootstrap replicates generate mobility candidates, and 2,000 validation state-path-group bootstrap replicates select the smallest set covering at least 95% of predictive-likelihood winners.

The validation winner distribution is concentrated on two development-only smoothed population-prior candidates, with selection frequencies 84.15% and 12.80%. Retaining these two models gives 96.95% cumulative validation selection mass, above the prespecified 95% target. The development MLE is 0.0514 with

a 95% bootstrap interval 0.0298–0.0754; the validation point estimate is 0.0817. Availability remains externally specified because GeoLife contains trajectories rather than task-delivery, willingness, or deadline observations.

A validation-best single-model BSP control is retained to separate improved model fit from finite-set robustness. The strict data-driven R-BSP and the disclosure-floor-aware RDF-BSP service frontier are reported in the main manuscript rather than duplicated here.

4. Reproducibility record

The repository stores deterministic preprocessing, split metadata, configuration files, seeds, trajectory-level summaries, bootstrap outputs, provenance records, analysis code, and manuscript sources. The effective GeoLife analysis split contains 23 development, 12 validation, and 53 test windows after model-input-group isolation. User IDs and complete 48-slot discrete state-path fingerprints are disjoint across the effective splits.

GeoLife point estimates average effective user windows and their constituent positions, whereas uncertainty intervals resample complete state-path groups so that duplicate model-input paths are not treated as independent clusters. The current automated suite contains 31 tests covering nominal, robust, and disclosure-floor posterior branches; gate reductions and maximality; informed-attacker integration; transition conservation; clustered-estimator weighting; GeoLife split and missing-minute handling; numerical replay; PML linear-program checks; PRIVIC author-code checks; and policy-envelope optimization.

The data-driven replay is reported as exploratory because the test replay had been inspected in preceding analyses; it is not treated as independent confirmatory validation. No mobile-device energy measurement, production MCS task-log study,

or independent external reproduction is claimed. The original Microsoft GeoLife archive and third-party article full texts are not redistributed in the submission package.

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